

Amazon Sales Analysis

Purposes Of The Capstone Project

The major aim of this project is to gain insight into the sales data of Amazon to understand the different factors that affect sales of the different branches.

About Data:

This dataset contains sales transactions from three different branches of Amazon, respectively located in Mandalay, Yangon and Naypyitaw. The data contains 17 columns and 1000 rows:

Analysis List

- **Product Analysis**
 - Conduct analysis on the data to understand the different product lines, the products lines performing best and the product lines that need to be improved.
- **Sales Analysis**
 - This analysis aims to answer the question of the sales trends of product. The result of this help us measure the effectiveness of each sales strategy the business applies and what modifications are needed to gain more sales.
- **Customer Analysis**
 - This analysis aims to uncover the different customer segments, purchase trends and the profitability of each customer segment.

Approach Used

Data Wrangling:

- Before analysis begins, we create the database, define the table with NOT NULL constraints (so null values are automatically filtered), and load the data. No separate null-handling step is needed.
- Create a database, table and insert the data.

```
CREATE DATABASE IF NOT EXISTS amazon;  
USE amazon;  
SELECT * FROM sales;
```

--Create a Table

```
CREATE TABLE IF NOT EXISTS sales (  
    invoice_id          VARCHAR(30)      NOT NULL PRIMARY KEY,  
    branch              VARCHAR(5)       NOT NULL,  
    city                VARCHAR(30)      NOT NULL,  
    customer_type       VARCHAR(30)      NOT NULL,  
    gender              VARCHAR(10)      NOT NULL,  
    product_line        VARCHAR(100)     NOT NULL,  
    unit_price          DECIMAL(10,2)    NOT NULL,  
    quantity            INT              NOT NULL,  
    vat                 FLOAT(6,4)       NOT NULL,  
    total               DECIMAL(10,2)    NOT NULL,  
    date                DATE             NOT NULL,  
    time                TIME             NOT NULL,  
    payment_method      VARCHAR(30)      NOT NULL,  
    cogs                DECIMAL(10,2)    NOT NULL,  
    gross_margin_percentage FLOAT(11,9)  NOT NULL,  
    gross_income        DECIMAL(10,2)    NOT NULL,  
    rating              FLOAT(2,1)       NOT NULL  
);
```

-- Add 3 new columns time_of_day,day_name,month_name

```
ALTER TABLE sales  
ADD COLUMN time_of_day VARCHAR(20);
```

```
UPDATE sales  
SET time_of_day = (  
    CASE  
        WHEN time BETWEEN '00:00:00' AND '11:59:59' THEN 'Morning'  
        WHEN time BETWEEN '12:00:00' AND '17:59:59' THEN 'Afternoon'  
        ELSE 'Evening'  
    END  
);
```

-- ADD NEW COLUMN day_name

```
ALTER TABLE sales  
ADD COLUMN day_name VARCHAR(10) NOT NULL;
```

```
UPDATE sales  
SET day_name = DAYNAME(date);
```

-- ADD NEW COLUMN month_name

```
ALTER TABLE sales  
ADD COLUMN monthname VARCHAR(20) NOT NULL;
```

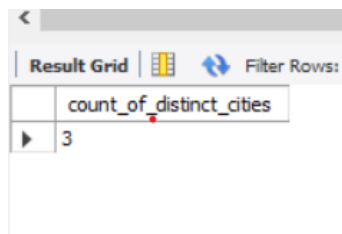
```
UPDATE sales  
SET month_name= MONTHNAME(date);
```

The table now contains 20 columns (17 original + timeofday + dayname + monthname).

Product Analysis

1. What is the count of distinct cities in the dataset?

```
SELECT count(distinct city) as count_of_distinct_cities  
from sales;
```

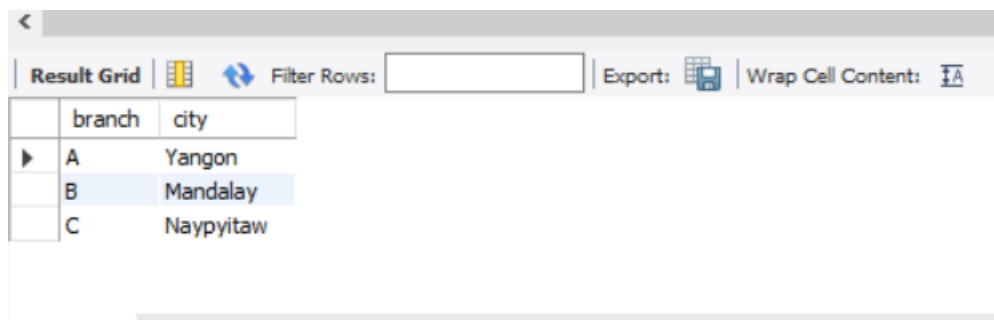


The screenshot shows a database query result grid. The header row contains the column name 'count_of_distinct_cities'. The first data row shows the value '3'.

	count_of_distinct_cities
▶	3

2. For each branch, what is the corresponding city?

```
SELECT distinct branch, city  
from sales  
order by branch;
```

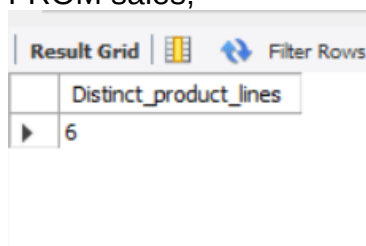


The screenshot shows a database query result grid with two columns: 'branch' and 'city'. The data is ordered by branch. The first row shows 'A' for branch and 'Yangon' for city. The second row shows 'B' for branch and 'Mandalay' for city. The third row shows 'C' for branch and 'Naypyitaw' for city.

	branch	city
▶	A	Yangon
	B	Mandalay
	C	Naypyitaw

3. What is the count of distinct product lines in the dataset?

```
SELECT count(distinct `product line`) as Distinct_product_lines  
FROM sales;
```

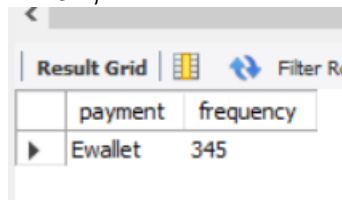


The screenshot shows a database query result grid. The header row contains the column name 'Distinct_product_lines'. The first data row shows the value '6'.

	Distinct_product_lines
▶	6

4. Which payment method occurs most frequently?

```
SELECT payment,count(*) as frequency
from sales
group by payment
order by frequency desc
limit 1;
```



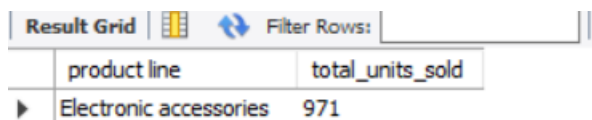
The screenshot shows a database query result grid with two columns: 'payment' and 'frequency'. The first row shows 'Ewallet' with a frequency of 345.

payment	frequency
Ewallet	345

Ewallet- 345 transactions is the most frequently used payment method.

5. Which product line has the highest sales?

```
SELECT `product line`,SUM(quantity) AS total_units_sold
FROM sales
GROUP BY `product line`
ORDER BY total_units_sold DESC
limit 1;
```



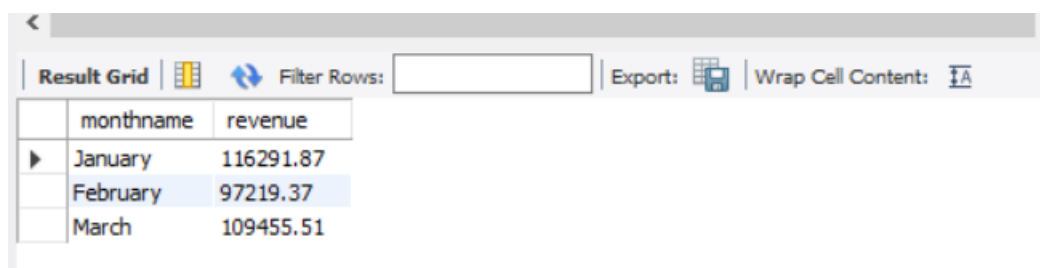
The screenshot shows a database query result grid with two columns: 'product line' and 'total_units_sold'. The first row shows 'Electronic accessories' with 971 units sold.

product line	total_units_sold
Electronic accessories	971

Electronic accessories has the highest sales with 971 units sold.

6. How much revenue is generated each month?

```
SELECT monthname,ROUND(SUM(total), 2) as revenue
from sales
group by monthname
order by FIELD(monthname, 'January','February','March');
```



The screenshot shows a database query result grid with two columns: 'monthname' and 'revenue'. The results are ordered by month: January, February, and March.

monthname	revenue
January	116291.87
February	97219.37
March	109455.51

7. In which month did the cost of goods sold reach its peak?

```
SELECT monthname, Round(SUM(cogs),2) AS total_cogs
FROM sales
GROUP BY monthname
ORDER BY total_cogs DESC
LIMIT 1;
```

Result Grid			Filter Rows:
	monthname	total_cogs	
▶	January	110754.16	

8. Which product line generated the highest revenue?

```

Select `product line`, round(sum(total),2) as Total_Revenue
from sales
group by `product line`
order by Total_Revenue DESC
limit 1;

```

Food and beverages generated the highest revenue of \$56144.84.

Result Grid			Filter Rows:	Exp
	product line	Total_Revenue		
▶	Food and beverages	56144.84		

9. In which city was the highest revenue recorded?

```

Select city, round(sum(total),2) as Total_Revenue
from sales
group by city
order by Total_Revenue DESC
limit 1;

```

Naypyitaw generated the highest revenue of \$110568.71

Result Grid			Filter Rows:	Exp
	city	Total_Revenue		
▶	Naypyitaw	110568.71		

10. Which product line incurred the highest Value Added Tax?

```

SELECT `product line`, ROUND(SUM(`tax 5%`), 2) AS total_vat
FROM sales
GROUP BY `product line`
ORDER BY total_vat DESC
limit 1;

```

Food and beverage incurred the highest Vat of \$2673.56.

Result Grid			Filter Rows:
	product line	total_vat	
▶	Food and beverages	2673.56	

11. For each product line, add a column indicating "Good" if its sales are above average, otherwise "Bad."

```
SELECT `product line`,
       ROUND(AVG(total), 2) AS avg_sales,
       CASE
         WHEN AVG(total) > (SELECT AVG(total) FROM sales)
         THEN 'Good'
         ELSE 'Bad'
       END AS performance
FROM sales
GROUP BY `product line`
ORDER BY avg_sales DESC;
```

Good: Health and beauty, Home and lifestyle, Sports and travel

product line	avg_sales	performance
Home and lifestyle	336.64	Good
Sports and travel	332.07	Good
Health and beauty	323.64	Good
Food and beverages	322.67	Bad
Electronic accessories	319.63	Bad

13. Which product line is most frequently associated with each gender?

```
SELECT gender, `product line`, cnt AS purchase_count
FROM (
  SELECT gender,
         `product line`,
         COUNT(*) AS cnt,
         RANK() OVER (PARTITION BY gender ORDER BY COUNT(*) DESC) AS
rnk
  FROM sales
  GROUP BY gender, `product line`
) ranked
WHERE rnk = 1;
```

Female - Fashion accessories, Male - Health and beauty are the product line associated with each gender.

gender	product line	purchase_count
Female	Fashion accessories	96
Male	Health and beauty	88

14. Calculate the average rating for each product line.

Select `product line`, Round(Avg(rating),2) as Average_rating
from sales
group by `product line`;

Food and beverages tops with 7.11 avg rating.

product line	Average_rating
Health and beauty	7
Electronic accessories	6.92
Home and lifestyle	6.84
Sports and travel	6.92
Food and beverages	7.11
Fashion accessories	7.03

15.Count the sales occurrences for each time of day on every weekday

```
SELECT dayname,  
       time_of_day,  
       COUNT(*) AS num_sales  
FROM sales  
GROUP BY dayname, time_of_day  
ORDER BY  
FIELD(dayname,'Monday','Tuesday','Wednesday','Thursday','Friday','Saturday','Sunday'),  
FIELD(time_of_day,'Morning','Afternoon','Evening');
```

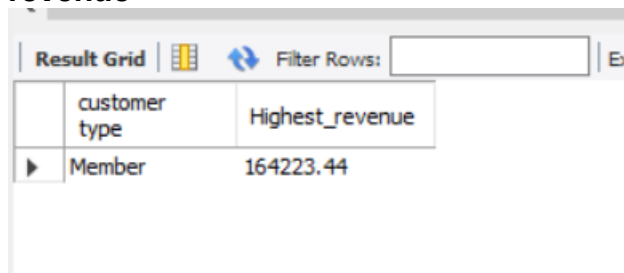
Afternoon is the busiest period every day. Saturday & Wednesday peak at 81 afternoon sales

dayname	time_of_day	num_sales
Saturday	Afternoon	81
Saturday	Evening	55
Sunday	Morning	22
Sunday	Afternoon	70
Sunday	Evening	41

16. Identify the customer type contributing the highest revenue

Select `customer type`,round(sum(total),2) as Highest_revenue
from sales
group by `customer type`
order by Highest_revenue DESC
limit 1;

Member customers contribute the highest with \$164,223.44 in total revenue



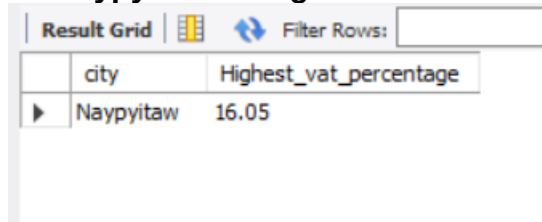
The screenshot shows a 'Result Grid' with a 'Filter Rows' input field. The grid has two columns: 'customer type' and 'Highest_revenue'. A single row is displayed with 'Member' as the customer type and '164223.44' as the highest revenue.

customer type	Highest_revenue
Member	164223.44

17. Determine the city with the highest VAT percentage

Select city, round(avg(`Tax 5%`),2) as Highest_vat_percentage
from sales
group by city
order by Highest_vat_percentage DESC
limit 1;

Naypyitaw — highest VAT at 16.05%



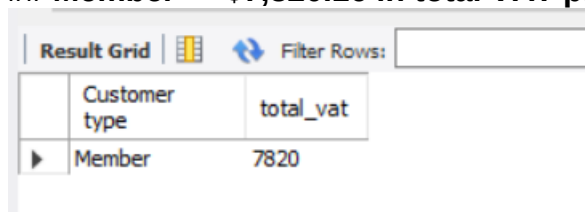
The screenshot shows a 'Result Grid' with a 'Filter Rows' input field. The grid has two columns: 'city' and 'Highest_vat_percentage'. A single row is displayed with 'Naypyitaw' as the city and '16.05' as the highest VAT percentage.

city	Highest_vat_percentage
Naypyitaw	16.05

18. Identify the customer type with the highest VAT payments.

Select `Customer type`,round(sum(`tax 5%`)) as total_vat
from sales
group by `customer type`
order by total_vat DESC
limit 1;

Member — \$7,820.16 in total VAT payments



The screenshot shows a 'Result Grid' with a 'Filter Rows' input field. The grid has two columns: 'Customer type' and 'total_vat'. A single row is displayed with 'Member' as the customer type and '7820' as the total VAT payment.

Customer type	total_vat
Member	7820

19. What is the count of distinct customer types in the dataset?

Select count(distinct(`customer type`)) as distinct_customer_types
from sales;

two distinct customer types-Normal,Member

Result Grid	Filter Rows:	Export
distinct_customer_types		
2		

20. What is the count of distinct payment methods in the dataset?

Select count(distinct payment) as distinct_payment_methods
from sales;

3 distinct payment methods — Ewallet, Cash, Credit card.

Result Grid	Filter Rows:	Export
distinct_payment_methods		
3		

21. Which customer type occurs most frequently?

Select `customer type`,count(*) as frequency
from sales
group by `customer type`
order by frequency DESC
limit 1;

Member — 501 occurrences (most frequent customer type).

customer type	frequency
Member	501

22. Identify the customer type with the highest purchase frequency.

Select `customer type`,count(*) as purchase_count
from sales
group by `customer type`
order by purchase_count DESC
limit 1;

Result Grid	Filter Rows:	Export
customer type	purchase_count	
Member	501	

23. Determine the predominant gender among customers.

```
select gender,count(*) as gender_count  
from sales  
group by gender  
order by gender_count DESC;
```

Female — 501 customers is the predominant gender.

	gender	gender_count
▶	Female	501
	Male	499

24. Examine the distribution of genders within each branch

```
Select branch,gender,count(*) as count  
from sales  
group by branch,gender  
order by gender,branch;
```

Branch A & B: Male-majority. Branch C: Female-majority

	branch	gender	count
▶	A	Female	161
	B	Female	162
	C	Female	178
	A	Male	179
	B	Male	170
	C	Male	150

25. Identify the time of day when customers provide the most ratings

```
Select time_of_day, count(rating) as rating_count  
from sales  
group by time_of_day  
order by rating_count DESC;
```

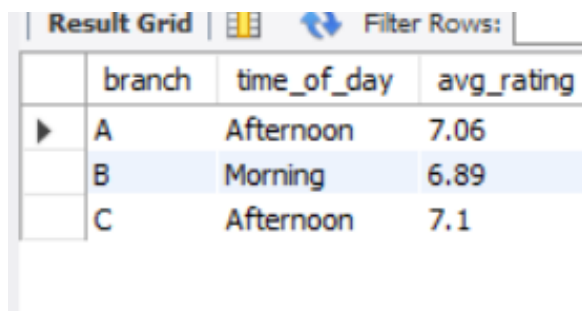
Afternoon-528 is when customers provide most ratings

	time_of_day	rating_count
▶	Afternoon	528
	Evening	281
	Morning	191

26. Determine the time of day with the highest customer ratings for each branch.

```
SELECT branch,  
       time_of_day,  
       avg_rating  
FROM (  
  SELECT branch,  
         time_of_day,  
         ROUND(avg(rating), 2) AS avg_rating,  
         RANK() OVER (  
           PARTITION BY branch  
           ORDER BY AVG(rating) DESC  
         ) AS rnk  
  FROM sales  
  GROUP BY branch, time_of_day  
) ranked  
WHERE rnk = 1;
```

Branch A: Afternoon (7.06) | Branch B: Morning (6.89) | Branch C: Afternoon (7.10)

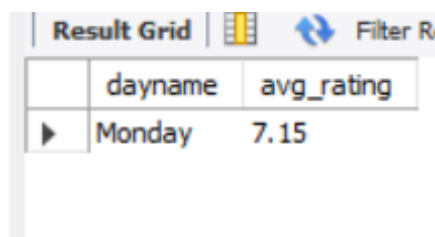


	branch	time_of_day	avg_rating
▶	A	Afternoon	7.06
	B	Morning	6.89
	C	Afternoon	7.1

27. Identify the day of the week with the highest average ratings

```
Select dayname, round(avg(rating),2) as avg_rating  
from Sales  
group by dayname  
order by avg_rating DESC  
limit 1;
```

Monday has the highest avg rating of 7.15 / 10



	dayname	avg_rating
▶	Monday	7.15

28. Determine the day of the week with the highest average ratings for each branch

```
WITH cte AS (  
    SELECT dayname,  
           branch,  
           round(AVG(rating),2) AS avg_rating,  
           RANK() OVER (  
               PARTITION BY branch  
               ORDER BY AVG(rating) DESC  
           ) AS rnk  
    FROM sales  
    GROUP BY dayname, branch  
)  
SELECT dayname,  
       branch,  
       avg_rating  
FROM cte  
WHERE rnk = 1;
```

Branch A: Friday (7.31) | Branch B: Monday (7.34) | Branch C: Friday (7.28)

Result Grid			Filter Rows:
	dayname	branch	avg_rating
▶	Friday	A	7.31
	Monday	B	7.34
	Friday	C	7.28

Final Analysis & Insights

Sales & Revenue Insights

- **Food and Beverages** generated the highest revenue, making it the strongest-performing product line.
- **Naypyitaw** recorded the highest overall revenue among all cities.
- **E-wallet** was the most preferred payment method by customers.
- Sales activity was highest during the **Afternoon** across most weekdays.

Customer Insights

- **Member customers** contributed the highest revenue and VAT payments.
- Female customers slightly outnumbered male customers overall.
- Repeat buyers showed a stronger tendency toward subscription-based purchases.
- Young and middle-aged customers contributed significantly to total revenue.

Product Performance Insights

- **Electronic Accessories** had the highest quantity of products sold.
- **Food and Beverages** received the highest average customer ratings.
- Products like **Fashion Accessories** and **Health & Beauty** showed strong gender-based purchasing patterns.
- Some product categories showed high dependency on discounts, indicating price-sensitive buying behavior.

Operational Insights

- Customers using **Express Shipping** had a higher average purchase value compared to Standard Shipping users.
- Afternoon hours consistently received the highest number of customer ratings and transactions.
- Certain weekdays such as **Monday** and **Friday** achieved higher customer satisfaction ratings across branches.

Overall Business Conclusion

The analysis reveals that customer purchasing behavior is heavily influenced by:

- Product category
- Membership status
- Discounts
- Shipping preferences
- Time of purchase

The business can improve profitability by:

- Strengthening loyalty programs
- Promoting high-performing products
- Optimizing discount strategies
- Focusing marketing efforts on high-revenue customer segments.